A Unified Bayesian Collective Reserving Framework with Hierarchical Temporal Smoothing

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Abstract

This paper develops a unified Bayesian reserving framework built upon the foundations of the Collective Reserving Model (CRM). Rather than proposing a single fixed specification, we introduce a modular probabilistic architecture that enables the principled integration of structural components commonly encountered in actuarial practice. These include hierarchical partial pooling across claim types, smooth temporal dynamics, spline-based development patterns, calendar effects, seasonal components, reporting–payment delay interactions, and portfolio-level risk covariates.

By utilizing Hamiltonian Monte Carlo implemented in `Stan`, all framework components are estimated jointly within a single coherent likelihood, allowing for a separate estimation of the RBNS and IBNR reserves. This unified formulation ensures full propagation of parameter and process uncertainty across modeling layers and yields robust posterior predictive distributions for reserve risk assessment.

Applied to a large European automobile physical damage portfolio, the framework demonstrates a strong ability to extract stable, high-resolution signals from granular monthly data. A rolling-origin backtest shows that the proposed approach substantially outperforms traditional deterministic benchmarks. Ultimately, the results demonstrate that a structured Bayesian approach provides a flexible, interpretable, and practically effective alternative to traditional reserving models.

1 Introduction

In Property and Casualty (P&C) insurance, accurate loss reserving plays a central role in ensuring financial stability, meeting regulatory requirements, and supporting effective risk management. Reserves are typically decomposed into two components: Reported But Not Settled (RBNS) claims and Incurred But Not Reported (IBNR) claims. Traditionally, reserving has been performed using *macro-level* methods that operate on highly aggregated claims data, most commonly organized into run-off triangles by accident and development periods. Prominent examples include the Chain Ladder method (Mack (1993, 1999)) and the Bornhuetter–Ferguson method (Bornhuetter and Ferguson (1972)), which model cumulative or incremental payments at the triangle level rather than individual claims or reporting cohorts. Although appealing for their simplicity and transparency, such aggregation obscures important claim-level dynamics and results in a small number of observations, leading to substantial parameter uncertainty and potential bias in certain reserving environments (see Charpentier et al. (2016)). Moreover, classical macro-level approaches do not permit separate estimation of RBNS and IBNR reserves.

Beyond these limitations, highly aggregated triangle structures also prevent models from exploiting important temporal and portfolio-level signals present in modern insurance datasets. In many contemporary reserving environments, claims are recorded at much finer temporal resolutions (e.g., daily), and portfolios often exhibit meaningful variation across time, claim types, and settlement dynamics. Working at finer resolutions allows models to capture such signals directly and to align

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reserve estimation with operational decision-making processes. However, moving to granular resolutions substantially increases the dimensionality of the reserving problem and leads to extreme sparsity in many development cells, creating significant statistical challenges for traditional modeling approaches.

The Collective Reserving Model (CRM), introduced by Wahl et al. (2019), enhances traditional reserving by conditioning on reported claim counts and modeling payments at a finer level of granularity, thereby enabling structural separation of RBNS and IBNR components. One used specification, described in Lindholm et al. (2020), is given by

$$X_{i,j,k} \mid N_{i,j} \sim \mathrm{ODP}\big(\psi_{i,j,k} N_{i,j}, \phi\big), \qquad N_{i,j} \sim \mathrm{ODP}(\nu_{i,j}, \varphi),$$

so that

$$\mathbb{E}[X_{i,j,k} \mid N_{i,j}] = \psi_{i,j,k} N_{i,j}, \qquad \mathbb{E}[N_{i,j}] = \nu_{i,j},$$

where ODP denotes the over-dispersed Poisson distribution. Here, $X_{i,j,k}$ denotes total payments arising from claims occurring in accident period i, reported with delay j, and paid k periods after reporting, while $N_{i,j}$ represents the number of reported claims in cell (i, j). Conditional on the claim counts, all payment amounts $X_{i,j,k}$ are assumed independent, and the claim counts $N_{i,j}$ are mutually independent.

Despite its conceptual appeal, practical implementation of the CRM presents substantial challenges. In particular, estimation of the cell-level parameters $\psi_{i,j,k}$ is complicated by severe overparameterization. Restrictive specifications such as $\psi_{i,j,k} = \gamma_k$ sacrifice important heterogeneity, whereas more flexible multiplicative structures of the form $\psi_{i,j,k} = \alpha_i \beta_j \gamma_k$ introduce a large number of parameters that cannot be reliably estimated in sparse data settings. Treating these effects as independent fixed parameters further exacerbates instability by preventing information sharing across related cells, often resulting in volatile estimates and poor predictive performance.

These challenges become considerably more severe when the model is applied to granular reserving data. At monthly or higher temporal resolutions, the (i, j, k) grid expands dramatically while the number of observations within many cells becomes extremely small. As a result, traditional fixed-effect parameterizations become highly unstable and the estimation problem becomes severely ill-posed. Consequently, practical CRM implementations require modeling strategies that can share information across related cells while preserving important temporal and structural patterns in the data.

To address these issues, machine learning techniques—including gradient boosting methods and neural networks—have been proposed to learn flexible functional representations of the form $\psi_{i,j,k} = f(i, j, k; \theta)$ (e.g. Lindholm et al. (2020) and Delong et al. (2022)). While such approaches mitigate the flexibility–estimability trade-off, they typically operate as black-box models with limited interpretability, provide incomplete uncertainty quantification, and may overfit in the presence of sparse historical data.

An alternative approach is to explicitly model the underlying stochastic mechanisms driving the claim development process using a fully probabilistic formulation. Such generative models describe how observed data arise from latent claim processes and allow future claim trajectories to be simulated directly from the model. This perspective provides several advantages in reserving applications: it enables coherent uncertainty quantification, preserves interpretability of model components, and allows domain knowledge about claim development dynamics to be incorporated in a principled manner.

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This paper develops a unified Bayesian framework for the CRM that addresses overparameterization while incorporating features of direct practical relevance. The proposed approach decomposes cell-level payment behavior into accident period, reporting delay, and development period components, together with their interactions. Rather than imposing restrictive multiplicative structures or estimating effects independently as fixed parameters, we employ structured temporal regularization through first-order random walks for accident and calendar year effects and spline-based representations for development patterns. These structures facilitate information sharing across adjacent time periods through both temporal dependence and basis representations.

A central feature of the proposed approach is that the entire reserving problem is formulated within a coherent probabilistic modeling framework implemented using modern probabilistic programming tools. In particular, we employ the probabilistic programming language `Stan` (Carpenter et al. (2017)), which enables sophisticated Bayesian inference using Hamiltonian Monte Carlo. This framework allows complex hierarchical structures, temporal dynamics, and interaction effects to be estimated jointly within a single model while maintaining computational stability and full uncertainty propagation.

When multiple claim types or business segments are present, the framework is extended hierarchically, allowing each group to retain its own temporal dynamics while borrowing strength through common hyperpriors. This hierarchical structure is particularly valuable under sparse data conditions, where fully separate models become unstable and complete pooling obscures genuine heterogeneity. Additional features commonly encountered in practical reserving—including seasonal patterns, interaction effects, and portfolio-level covariates—are incorporated modularly within the same probabilistic specification. All components are embedded within a coherent Bayesian framework, ensuring full uncertainty propagation across all modeling layers. Separate models are developed for RBNS and IBNR reserves, each yielding full posterior distributions suitable for risk assessment and capital analysis.

Importantly, the probabilistic programming framework also allows the model to be developed in a modular and extensible manner. Structural components such as hierarchical pooling, seasonal patterns, interaction effects, and portfolio-level signals can be incorporated incrementally within the same modeling architecture. This enables the analyst to progressively refine the model as additional signals are identified in the data, while maintaining a coherent probabilistic representation of the claim development process.

Bayesian methods have been applied extensively within the claims reserving literature. Early contributions include Bayesian formulations of the chain-ladder model developed by Verrall (1990), hierarchical Bayesian approaches to RBNS reserving based on joint modeling of claim counts and severities proposed by Ntzoufras and Dellaportas (2002), and Bayesian over-dispersed Poisson models incorporating exposure measures and inflation effects studied by Scollnik (2004). Subsequent work expanded the scope of Bayesian reserving models by addressing practical data challenges. De Alba (2002) introduced Bayesian techniques for handling negative incremental claims and incorporating actuarial judgment through prior distributions, while Meyers (2015) developed MCMC-based chain-ladder models with calendar year effects and demonstrated their performance using the CAS Loss Reserve Database. More recent contributions have focused on improving flexibility and stability under complex or sparse data environments. These include hierarchical Bayesian compartmental reserving models implemented in Stan (Gesmann and Morris (2020)), spline-based approaches for modeling non-linear development patterns (Gao and Meng (2018)), and Bayesian MCMC frameworks designed to stabilize reserve estimates in the presence of noise and limited historical information (Larsen (2024)).

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While these contributions demonstrate the value of Bayesian inference for improving estimation stability and uncertainty quantification in aggregated reserving models, they remain primarily formulated at the macro-level of run-off triangles. Consequently, they do not provide an explicit probabilistic representation of the underlying claim process, nor do they offer a unified modeling structure for jointly incorporating multiple sources of temporal variation, heterogeneity, and dependence within a single reserving framework. The present work extends this literature by developing a unified Bayesian formulation of the CRM, in which temporal smoothing, hierarchical pooling across claim types, interaction structures, seasonal effects, calendar year adjustments, and portfolio-level covariates are integrated consistently within one probabilistic architecture, enabling stable and interpretable estimation of RBNS and IBNR reserves.

The proposed framework is illustrated using a large automobile physical damage portfolio from a European insurer, where we conduct our analysis at a monthly level of granularity. Model estimation is carried out using Hamiltonian Monte Carlo implemented in `Stan`, enabling full Bayesian inference with automated convergence diagnostics. To assess predictive performance under realistic reserving conditions, reserve estimates are produced for twelve consecutive valuation dates and validated against subsequent observed claim payments. Through this empirical study, we demonstrate how predictive accuracy and uncertainty quantification evolve as structural components are progressively incorporated into the model. In particular, we show how temporal smoothing, hierarchical pooling, seasonal effects, and interaction structures contribute to substantial improvements in reserve stability and out-of-sample performance. Complete reproducible `Stan` code is provided to facilitate practical implementation and to support adoption of the proposed framework by practitioners.

This paper builds on our earlier work in Abdelrahman et al. (2025), which studies Bayesian temporal modeling for IBNR claim estimation and demonstrates how appropriately structured temporal effects can capture key empirical features of the data while outperforming substantially more complex models proposed in the literature. While that work focuses on illustrating the effectiveness of flexible temporal structures for predictive accuracy in claim count modeling, the present paper applies the same modeling philosophy within the CRM framework to address the full reserving problem.

The remainder of this paper is organized as follows. Section 2 describes the European automobile insurance dataset and presents an exploratory analysis of reporting and settlement patterns. Section 3 develops the Bayesian payment modeling framework used for RBNS reserve estimation, detailing the hurdle Log-normal specification and hierarchical extensions. Section 4 formulates the IBNR component, focusing on the joint modeling of claim counts and type composition. Section 5 presents the results of a rolling backtest, evaluating the proposed framework against traditional benchmarks. Finally, we conclude in Section 6.

2 Data Description and Exploratory Analysis

Before developing the modeling framework, we describe the dataset and present key empirical features of the claims process that inform the subsequent model design. The purpose of this section is descriptive: we document reporting, payment, and settlement patterns observed in the data without imposing a parametric structure. In the following sections, these empirical findings are translated into a formal Bayesian reserving model.

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2.1 Dataset Description

The dataset comprises auto insurance Physical Damage policies from a European insurer, previously examined in Abdelrahman et al. (2026). It includes policies issued between January 2009 and December 2017, covering a total of 293,709 one-year policies and 106,187 reported claims. Each policy record contains policyholder and vehicle characteristics, while claim records include the date of occurrence, reporting date, and the complete payment history observed through the end of 2017.

For model development and validation, we use claims occurring before January 1, 2015, to estimate reserve quantities. Reserve estimates are computed at twelve valuation dates corresponding to the end of each month from January 2014 through December 2014. These projections are validated against actual claim payments observed through December 31, 2017. This three-year validation window corresponds to the maximum development horizon in the data, as virtually all claims are fully settled within this period. The design reflects a realistic reserving exercise in which future payments must be projected based on information available at each valuation date.

Throughout the analysis, reporting delays and settlement delays are measured in discrete monthly intervals. A claim occurring in month t and reported in the same month has reporting delay $d = 0$, while a claim reported in month $t + 1$ has $d = 1$, regardless of the exact number of days elapsed. Similarly, payment delays are measured as the number of months between reporting and payment.

2.2 Temporal Patterns in Claim Frequency & Severity

We first examine temporal patterns in claim frequency and severity. The data exhibit pronounced temporal variation in claim frequency. Although exposure, measured by the number of active policies, follows a clear pattern—rising steadily until late 2012, declining through early 2015, and subsequently increasing again—the corresponding claim frequency per unit of exposure displays more irregular behavior. Over the observation period, claim frequency varies between 0.023 and 0.045 claims per exposure. Figure 1, reproduced from Abdelrahman et al. (2026), illustrates the evolution of both exposure and claim frequency per exposure over time.

Figure 2 presents the temporal evolution of mean losses per claim for each accident period, computed as the ratio of total payments to the number of reported claims. The series displays substantial variability across periods, with several pronounced peaks indicating the presence of high-severity claims in specific accident months. Although some short-term fluctuations appear visually clustered over time, these patterns are not strictly periodic and vary in both magnitude and persistence across years.

2.3 Analysis of Severity

To further examine the distributional characteristics of claim severity, we consider the total loss per claim, defined as the sum of all payments associated with an individual claim. Figure 3 displays the distribution of total claim losses on a logarithmic scale. The distribution is strongly right-skewed, a pattern characteristic of insurance severity data. The logarithmic transformation substantially stabilizes the distribution, yielding a smooth and unimodal shape.

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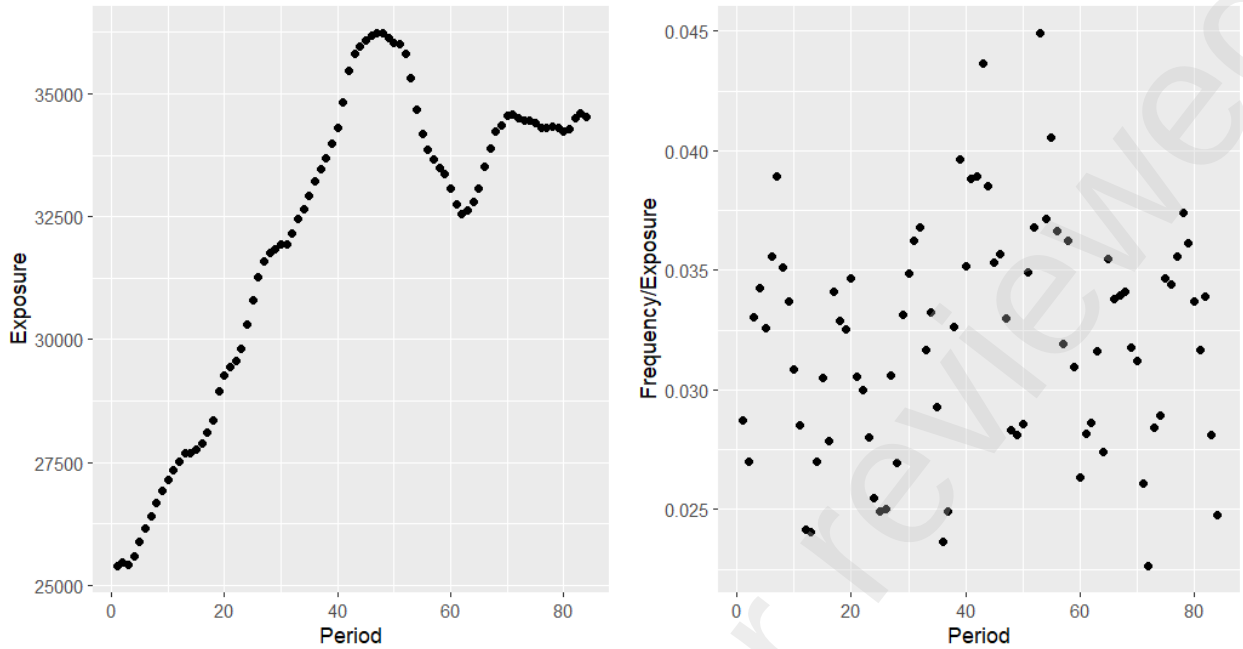

Figure (1) Exposure & Claim Count Per Exposure Between Jan 2009 and Dec 2015 (Reproduced from Abdelrahman et al. (2026))

2.4 Reporting Delay Patterns

We next examine reporting delays, defined as the time in months between claim occurrence and initial reporting. Table 1 presents the empirical distribution of reporting delays for claims in the development sample.

Table (1) Distribution of Reporting Delays (in months)

Delay (d)	0	1	2	3	4	5	6	7	8	9	10+	Total
Claims	57,411	10,045	1,317	552	266	166	106	84	68	37	107	70,159
%	81.8	14.3	1.9	0.8	0.4	0.2	0.2	0.1	0.1	0.1	0.2	100.0
Cum. %	81.8	96.1	98.0	98.8	99.2	99.4	99.6	99.7	99.8	99.9	100.0	

The data exhibit rapid reporting behavior typical of auto physical damage claims. Approximately 82% of claims are reported in the same month as occurrence, and over 99% are reported within three months. All claims are reported within two years of occurrence.

2.5 Payment Patterns

We next examine payment delays, defined as the time in months between reporting and individual claim payments. Table 2 presents the distribution of payment delays along with summary statistics of payment amounts.

Table (2) Distribution of Payment Delays and Payment Characteristics (in months)

Delay (s)	0	1	2	3	4	5	6	7	8	9	10	11	12+
Payments	8,443	44,551	28,814	7,604	2,800	1,441	975	599	514	372	321	253	1,623
%	8.6	45.3	29.3	7.7	2.9	1.5	1.0	0.6	0.5	0.4	0.3	0.3	1.7
Cum. %	8.6	53.9	83.2	90.9	93.8	95.3	96.3	96.9	97.4	97.8	98.1	98.3	100.0
Mean	2,571	2,728	2,706	3,484	3,584	3,151	2,810	2,417	2,989	2,068	2,328	1,792	2,392

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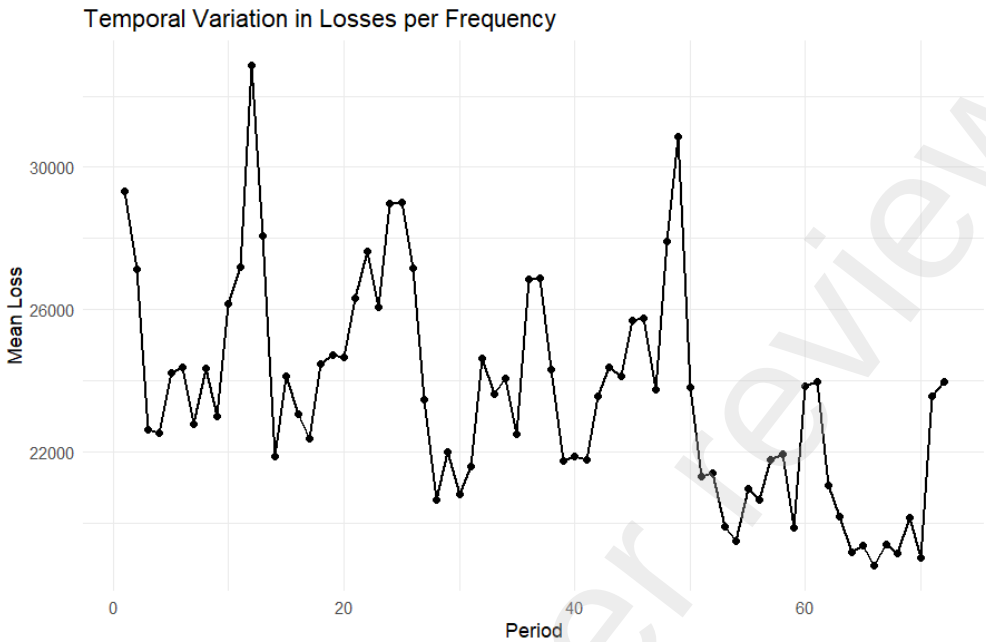

Figure (2) Mean Loss per Claim Between Jan 2009 and Dec 2015

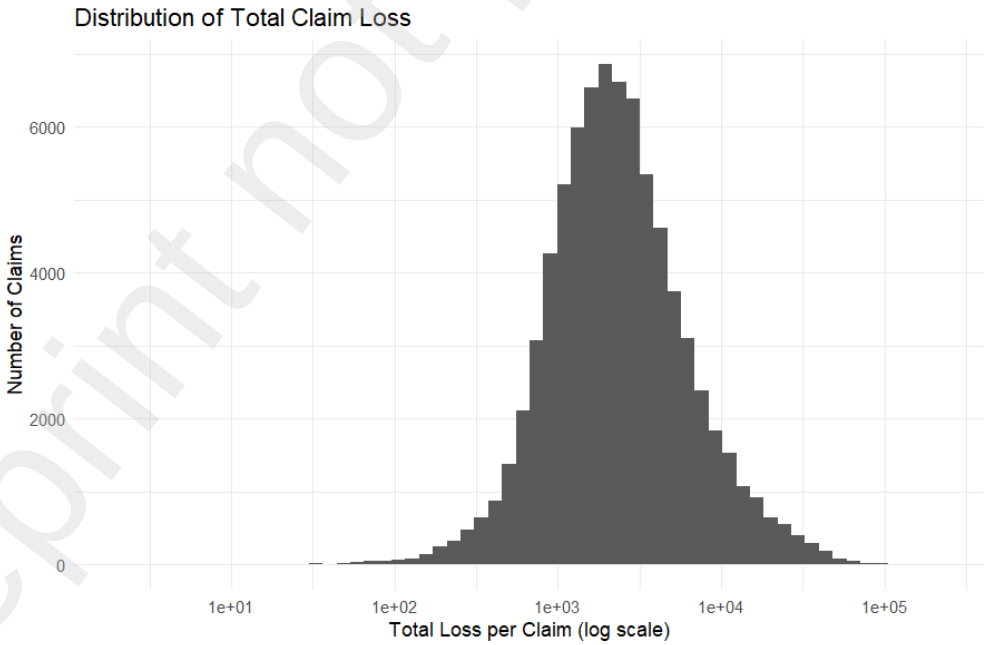

Figure (3) Distribution of Claim Severity

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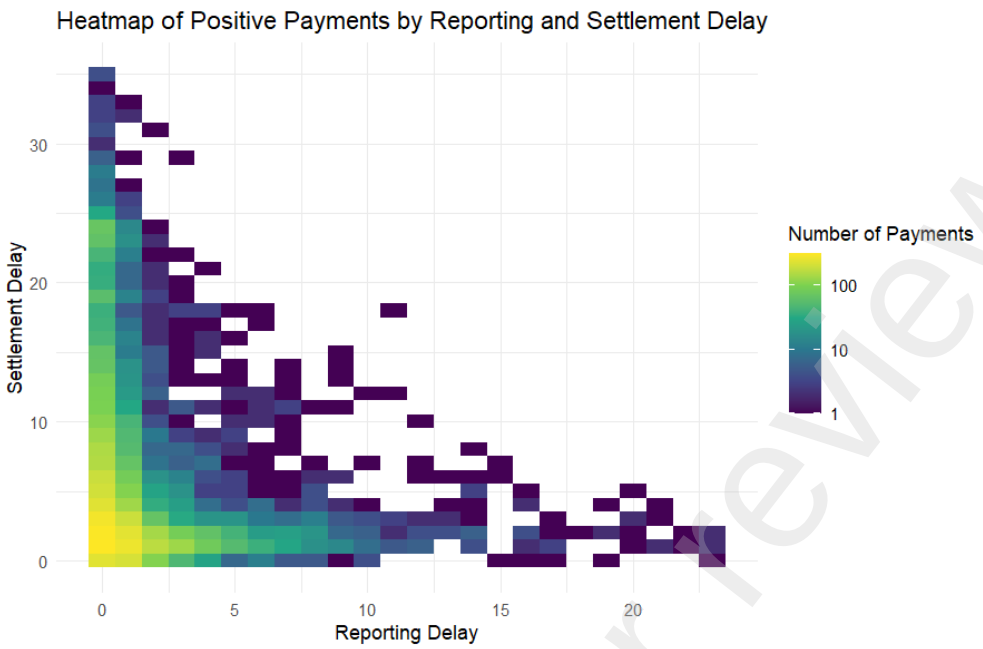

Figure (4) Heatmap of the Number of Positive Payments across Reporting and settlement Delays

Payments are strongly front-loaded: over half occur within one month of reporting, and more than 83% occur within the first two months. However, a meaningful fraction of payments (16.8%) occurs beyond two months, with 1.7% extending beyond eleven months. Mean payment amounts vary systematically across settlement delays. Payments occurring at delays $s = 3$–5 exhibit higher average severities than those settled immediately, suggesting more complex claims or multi-stage settlement processes. Later payments display greater variability, reflecting heterogeneous claim resolution paths.

Although the reserving model does not explicitly condition on the number of payments per claim, it is worth noting that 72.9% of the claims are settled with a single payment, while 27.1% involve multiple payments. Nearly 91% of claims require at most two payments, consistent with the operational characteristics of physical damage claims.

Figure 4 displays a heatmap of the number of positive payments across reporting delays and settlement delays. The distribution is highly concentrated in cells with short reporting and development periods, while observations become increasingly sparse as either dimension increases. In particular, cells corresponding to long reporting delays and extended settlement periods contain very few payments, highlighting substantial sparsity in the payment data.

The exploratory analysis highlights several salient features of the claims process: rapid but imperfect reporting, strongly front-loaded payments with a non-negligible tail, systematic variation in payment severity across settlement delays, and substantial sparsity in late-delay cells. These characteristics underscore the limitations of fixed-effect cell-level modeling and motivate the need for structured regularization, partial pooling, and flexible temporal effects. In the next section, we develop a Bayesian RBNS reserving framework that directly addresses these empirical features.

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3 RBNS Reserving Framework

In this section, we develop a Bayesian framework for estimating RBNS reserves. To this end, we construct a probabilistic model for the payment process, and the RBNS reserve is obtained as the posterior expectation of future payments conditional on reported claims. The model builds on the CRM of Wahl et al. (2019) and Lindholm et al. (2020), and is explicitly designed to accommodate the empirical patterns documented in Section 2. In particular, the framework addresses front-loaded yet heterogeneous payment dynamics, sparsity across reporting and settlement delays, and temporal variation observed at a monthly resolution.

3.1 Notation and Data Structure

Let $t \in \{1, \ldots, T\}$ denote the accident period, $d \in \{0, \ldots, D\}$ the reporting delay, and $s \in \{0, \ldots, S\}$ the payment delay. We define $N_{t,d}$ as the number of claims occurring in period t and reported in period $t + d$, and $X_{t,d,s}$ as the total payments made in period $t + d + s$ for those claims.

Consistent with the empirical setup in Section 2, all time indices are defined at a monthly resolution. This choice allows the model to capture intra-year dynamics, incorporate seasonality, and align valuation dates with operational reserving practices.

While the classical CRM assumes an over-dispersed Poisson distribution for both claim counts and payments, we relax this assumption for the payment process. This flexibility allows us to explore alternative likelihoods that better reflect the semi-continuous, right-skewed nature of insurance payments observed in the data.

3.2 Reserve Definition and Valuation

At valuation date T, observed data consist of claim counts $N_{t,d}$ with $t+d \leq T$ and payments $X_{t,d,s}$ with $t+d+s \leq T$. The RBNS reserve corresponds to future payments for claims that have already been reported but are not yet fully settled. The aggregate RBNS reserve is defined as

$$\hat{R}^{RBNS} = \sum_{t+d \leq T} \sum_{t+d+s > T} \mathbb{E}[X_{t,d,s} \mid \mathcal{D}_T], \tag{1}$$

where $\mathcal{D}_T$ denotes all information available at time T.

3.3 Sequential Model Development and Validation

We develop the payment model in a sequential manner, starting from a parsimonious baseline specification and progressively incorporating additional structural components, including calendar year effects, seasonal patterns, and selected interaction terms. This stepwise construction is designed to serve both practitioners and academic researchers by illustrating how model complexity may be expanded in a controlled and interpretable fashion, guided by data availability, empirical diagnostics, and domain expertise.

Importantly, the inclusion of certain components is not solely driven by statistical significance within the present dataset. Instead, we deliberately consider a comprehensive set of modeling features commonly encountered in insurance reserving applications, even when their empirical support is weak or inconclusive in this particular case study. This design choice ensures that the proposed framework remains flexible and broadly applicable across lines of business, portfolios, and data environments with differing structural characteristics.

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Throughout the sequential development, each model extension is assessed relative to the preceding specification to examine how additional structure affects RBNS reserve estimates and predictive stability under a rolling backtest design. Reserve estimates are computed for twelve valuation dates corresponding to the end of each month in 2014 and are validated against observed payments through December 31, 2017. Predictive accuracy is assessed using the Mean Absolute Percentage Error (MAPE),

$$\text{MAPE} = \frac{1}{12} \sum_{v=1}^{12} \left| \frac{R_v^{\text{actual}} - \hat{R}_v^{\text{pred}}}{R_v^{\text{actual}}} \right|, \tag{2}$$

where R_v^{actual} and $\hat{R}_v^{\text{pred}}$ denote the realized and predicted RBNS reserves for valuation date v, respectively. Results for the sequential specifications are reported in Section 5.

We implement our models using the `Stan` probabilistic programming language, which utilizes Hamiltonian Monte Carlo (HMC) to generate posterior samples for all model parameters. These samples are then used to generate the posterior predictive distribution of the reserve by simulating future payment trajectories. Given the characteristic right-skewness of insurance loss distributions, we adopt the posterior median as our primary point estimate for the reserve to ensure robustness against extreme tail events. Furthermore, the Bayesian approach allows us to construct predictive intervals directly from the posterior samples, facilitating the rigorous uncertainty quantification and risk assessment presented in the following sections.

3.4 The Baseline Payment Model: A Hurdle Approach

The primary challenge in modeling cell-level payments $X_{t,d,s}$ arises from the empirical structure of the payment data documented in Section 2. In particular, the payment process exhibits a substantial number of zero-valued observations across reporting and settlement delays, together with a continuous and strongly right-skewed distribution for strictly positive amounts. This semi-continuous behavior is especially pronounced in cells corresponding to longer reporting delays and extended settlement periods, where observations become increasingly sparse.

To accommodate these features, we employ a Hurdle Log-normal model, which decomposes the payment process into two independent components: a Bernoulli model governing the occurrence of payments and a Log-normal model describing the magnitude of positive payments. This formulation allows the model to explicitly separate the mechanisms driving payment incidence from those governing payment severity, which is particularly important in settings characterized by heterogeneous development patterns and excess zeros.

While the Tweedie distribution is a common alternative for modeling semi-continuous insurance data, its direct implementation within the HMC framework of `Stan` is not straightforward. For this reason, we focus on hurdle-type specifications, which provide comparable modeling flexibility while enabling full Bayesian inference through HMC.

Several candidate likelihoods were evaluated within the hurdle framework, including a Gamma-hurdle specification as well as the traditional over-dispersed Poisson (ODP) formulation used in classical reserving model. Among these alternatives, the hurdle Log-normal specification achieved superior empirical fit, as measured by the marginal likelihood and out-of-sample predictive performance. Consequently, the Log-normal severity model is adopted as the baseline specification in the subsequent analysis.

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Structural Specification

Let $Z_{t,d,s}$ denote a binary indicator such that $Z_{t,d,s} = 1$ if $X_{t,d,s} > 0$ and $Z_{t,d,s} = 0$ otherwise. In words, $Z_{t,d,s} = 1$ indicates that a positive payment occurs in calendar period $t + d + s$ for claims from accident period t reported with delay d, and $Z_{t,d,s} = 0$ otherwise. The conditional mean payment parameter $\psi_{t,d,s}$ can be expressed as

$$\psi_{t,d,s} = \mathbb{P}(Z_{t,d,s} = 1 \mid N_{t,d}) \times \mathbb{E}[X_{t,d,s} \mid Z_{t,d,s} = 1,\, N_{t,d}]. \tag{3}$$

We specify separate linear predictors for the payment probability and severity components:

$$\eta^{\text{prob}}_{t,d,s} := \text{logit}(\mathbb{P}(Z_{t,d,s} = 1)) = \mu^{\text{prob}} + \alpha^{\text{prob}}_t + \beta^{\text{prob}}_d + \zeta^{\text{prob}}_s + \theta \log N_{t,d}, \tag{4}$$

$$\eta^{\text{sev}}_{t,d,s} := \log(\mathbb{E}[X_{t,d,s} \mid Z = 1]) = \mu^{\text{sev}} + \alpha^{\text{sev}}_t + \beta^{\text{sev}}_d + \zeta^{\text{sev}}_s + \log N_{t,d}. \tag{5}$$

Here, μ denotes a global intercept. The terms α_t, β_d, and ζ_s represent accident-period, reporting-delay, and payment-delay (settlement) effects, respectively. Superscripts "prob" and "sev" indicate that separate sets of these effects are estimated for the payment-incidence and payment-severity components.

In the severity equation, $\log N_{t,d}$ acts as an offset to ensure proportionality with claim volume, while the coefficient θ in the probability equation allows for a flexible relationship between claim volume and the likelihood of observing at least one payment within a cell.

Prior Structure

Unless otherwise stated, the prior structures described below are applied separately to both the payment probability and severity components. For notational simplicity, we omit superscripts in this subsection; each prior specification should be understood as applying to both sets of parameters.

To ensure stable estimation across the (t, d, s) grid, we adopt weakly informative priors together with non-centered parameterizations, motivated by the empirical characteristics of the data.

- **Accident Period Effects (α_t):** To capture temporal variation in payment levels, as illustrated in Figure 2, accident-period effects are modeled using a first-order random walk. This specification acts as a temporal smoother, allowing gradual evolution over time while avoiding sensitivity to short-term noise. A non-centered parameterization is employed:

  $$\alpha_t = \alpha_{t-1} + \sigma_\alpha\, \alpha_{\text{raw},t}, \qquad \alpha_{\text{raw},t} \sim \text{Normal}(0, 1), \quad \sigma_\alpha \sim \text{Half-Normal}(0, 1).$$

  This formulation is mathematically equivalent to the standard (centered) random-walk specification in which the increment is modeled directly as $\varepsilon_t \sim \text{Normal}(0, \sigma^2_\alpha)$. The non-centered representation does not change the statistical model or its interpretation; it is adopted purely for computational reasons. In hierarchical Bayesian models, variance parameters such as σ_α can exhibit strong posterior dependence with the latent states under a centered parameterization, which may slow Markov chain Monte Carlo sampling. The non-centered form reduces this dependence and improves sampling stability while preserving the same probabilistic structure.

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- **Reporting Delay Effects (β_d):** Given the strong concentration of early reporting delays and the extreme sparsity observed for longer delays (Table 1), reporting delay effects are assigned hierarchical priors,

$$\beta_d \sim \text{Normal}(0, \sigma_\beta).$$

  The variance parameter σ_β is estimated from the data and governs the overall dispersion of reporting-delay effects. Rather than estimating each delay effect independently, the model allows them to vary around a common center with data-driven variability; this structure is known as partial pooling in Bayesian hierarchical modeling. As a result, well-populated delays are primarily informed by their own data, while sparse delays are automatically shrunk toward the overall mean, improving stability.

  For very long reporting delays, where observations become extremely rare, we introduce an additional structural regularization. Specifically, we cap the delay index at a fixed threshold $D_{\text{cap}} = 12$ and define

$$\beta_d = \beta_{D_{\text{cap}}} \quad \text{for all } d \geq D_{\text{cap}}.$$

  Thus, all reporting delays beyond 12 months share a common tail parameter. This deterministic pooling further stabilizes inference in the extreme tail of the reporting distribution.

- **Payment Delay Effects (ζ_s):** Payment development patterns documented in Section 2 exhibit strong front-loading, followed by increasingly sparse observations at longer settlement delays. To reflect this structure, we adopt a hybrid specification:

$$\zeta_s = \begin{cases} \zeta_s^{(\text{fixed})}, & s \leq 4, \\ \mathbf{B}_s^\top \mathbf{b}, & s > 4, \end{cases}$$

  where early settlement periods are modeled using independent fixed effects and later settlement periods are represented using a natural spline basis $\mathbf{B}_s$ with coefficient vector $\mathbf{b}$. The threshold $s = 4$ is motivated by the empirical payment distribution (see Table 2): approximately 95% of total paid losses occur within the first four settlement periods, while each subsequent period accounts for less than 1% of aggregate payments. Estimating unrestricted fixed effects beyond this point would therefore introduce parameters supported by extremely limited information.

  The spline representation enforces smooth evolution of the development pattern in the sparsely observed tail, preventing unstable month-to-month fluctuations driven by noise. The natural spline further stabilizes estimation by imposing linear behavior at the boundary of the development horizon. This structural regularization enforces smooth development behavior in the tail, consistent with the gradual settlement dynamics of physical damage claims.

Global intercepts are assigned weakly informative priors,

$$\mu^{\text{prob}} \sim \text{Normal}(-1, 2), \qquad \mu^{\text{sev}} \sim \text{Normal}(5, 2),$$

The prior mean -1 for the probability intercept places the baseline logit in a reasonable interior region, ensuring the implied payment probability lies between 0 and 1 without being pushed toward extreme values. The prior mean 5 for the severity intercept reflects the empirical scale of log-payments. The standard deviation of 2 keeps these intercept priors weakly informative.

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All auxiliary scale parameters (e.g., σ_α, σ_β, and spline-related scale terms) are assigned standard Half-Normal priors. These parameters control the degree of temporal smoothing and cross-cell variability, and the Half-Normal prior provides regularization while allowing substantial variation when supported by the data.

Overall, the proposed baseline framework directly addresses the overparameterization inherent in traditional triangular methods in which accident, reporting, and settlement effects are treated as independent fixed effects. By combining temporal smoothing, hierarchical pooling, and spline-based regularization, the model achieves stable and interpretable estimation across the full (t, d, s) grid while retaining sufficient flexibility to capture the empirical payment dynamics observed in the data.

This baseline hurdle Log-normal model is evaluated across the twelve valuation dates using the rolling backtest protocol described earlier in the Section. Its performance serves as a benchmark for assessing the incremental value of the subsequent structural extensions; quantitative results are reported in Section 5.

3.5 Hierarchical Extension: Partial Pooling Across Claim Types

While the baseline model captures global temporal and development patterns, it assumes a homogeneous population. However, empirical evidence suggests significant heterogeneity across different claim event causes. As shown in Table 3, the portfolio comprises multiple segments with vastly different frequency and severity profiles. The claim types correspond to the following categories: (1) Unknown party, (2) Known party, (3) Insured, (4) Natural cause, (5) Malfunction, and (6) Unspecified party (but identifiable).

Table (3) Distribution of Claims and Mean Loss by Claim Type

Claim Type	% of Total Claims	Mean Loss
1	37.80%	2,677
2	8.35%	5,791
3	52.40%	4,388
4	0.26%	5,376
5	0.01%	7,011
6	1.24%	5,317

The data reveals a critical modeling challenge: segments like Cause 5 and Cause 4 are extremely sparse, representing less than 0.3% of the total observations combined. Estimating independent models for these segments would lead to highly unstable parameters and unreliable reserve estimates. Conversely, a fully pooled model would ignore the distinct severity profile of Cause 5, which is nearly three times higher than Cause 1.

To navigate this trade-off, we extend the baseline model into a hierarchical framework. The key idea is to retain the same structural components introduced in Section 3 (accident-period, reporting-delay, and settlement effects), but to allow each claim type to deviate from a common portfolio-level pattern rather than estimating each type completely independently.

For each claim type $j \in \{1, \ldots, J\}$, the intercepts and temporal effects are modeled as group-specific deviations from a global mean. For example, the intercept for the severity component is defined as:

$$\mu_j^{\text{sev}} = \mu_0^{\text{sev}} + \sigma_{\mu,\text{sev}} \cdot \mu_{\text{raw},j}^{\text{sev}}$$

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Here, μ_0^{sev} denotes the global (portfolio-level) intercept shared across all claim types, while $\mu_{\text{raw},j}^{\text{sev}} \sim \text{Normal}(0,1)$ represents a standardized type-specific deviation. The scale parameter $\sigma_{\mu,\text{sev}}$ controls the magnitude of between-type variability. This formulation uses a non-centered parameterization to improve computational stability while preserving the same hierarchical structure.

The global intercept μ_0^{sev} retains the same prior specification as in Section 3, and $\sigma_{\mu,\text{sev}}$ is assigned a Half-Normal$(0,1)$ prior, consistent with the variance components introduced previously.

The hierarchical structure is applied to each structural component of the linear predictors in both the payment probability and severity models. For the accident-period effects, we allow each claim type to deviate from a global random-walk trajectory:

$$\alpha_{j,t} = \alpha_t^{\text{global}} + \sigma_{\alpha,\text{dev}} \cdot \delta_{j,t}, \qquad \delta_{j,t} \sim \text{Normal}(0,1),$$

where α_t^{global} follows the random-walk specification introduced in Section 3 and $\sigma_{\alpha,\text{dev}}$ controls the magnitude of cross-type deviations.

For the reporting-delay effects, we adopt a delay-specific hierarchical structure:

$$\beta_{j,d} = \beta_d^{\text{global}} + \sigma_{\beta,d} \cdot \epsilon_{j,d}, \qquad \epsilon_{j,d} \sim \text{Normal}(0,1),$$

where β_d^{global} denotes the global delay effect and $\sigma_{\beta,d}$ allows the degree of heterogeneity to vary across reporting delays.

For the settlement (payment-delay) effects, we retain the hybrid fixed–spline structure described in Section 3 and extend it hierarchically. Early settlement periods ($s \leq 4$) are modeled as

$$\zeta_{j,s}^{(\text{fixed})} = \zeta_{\text{global},s}^{(\text{fixed})} + \sigma_{\zeta,\text{fixed}} \cdot \kappa_{j,s}, \qquad \kappa_{j,s} \sim \text{Normal}(0,1),$$

while later settlement periods ($s > 4$) are represented using a natural spline basis:

$$\zeta_{j,s}^{(\text{spline})} = \mathbf{B}_s^\top \left(\mathbf{b}_{\text{global}} + \sigma_b \cdot \mathbf{b}_j^{\text{raw}}\right), \qquad \mathbf{b}_j^{\text{raw}} \sim \text{Normal}(\mathbf{0}, \mathbf{I}),$$

so that claim types deviate from a global settlement curve through deviations in spline coefficients rather than through independent month-to-month effects.

This hierarchical specification reflects three structural assumptions. First, claim types share a common underlying temporal evolution—i.e., the accident-period effect—captured by the global random walk, while allowing for regularized type-specific deviations. Second, reporting and settlement delay profiles are broadly similar across the portfolio, with deviations capturing systematic differences without requiring fully independent estimation for sparse types. Finally, the structure induces data-driven shrinkage: when a claim type has abundant data, its deviations are primarily informed by its own experience; when data are scarce, the posterior is pulled toward the global portfolio-level effects. This automatic borrowing of strength improves stability for low-volume claim types without eliminating genuine heterogeneity.

The hierarchical structure materially improves predictive stability by allowing claim-type-specific levels and trends while borrowing strength for sparse claim types through partial pooling. By differentiating claim causes in this manner, the model better reflects heterogeneous severity profiles without sacrificing calibration in low-volume segments. Quantitative comparisons under the rolling backtest design are provided in Section 5.

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3.6 Calendar Period Effect Extension

To account for exogenous shocks —such as legislative shifts, or inflationary pressure—we extended the severity component to include a calendar period effect, γ_τ, where $\tau = t + d + s$. We explored several specifications for this effect, comparing monthly versus quarterly resolutions, as well as global versus hierarchical structures across claim types. In all variations, γ_τ was modeled as a first-order random walk with a zero-mean constraint to ensure identifiability against the accident period and development delay effects.

Despite the theoretical justification for including a calendar-period index, adding γ_τ did not improve out-of-sample calibration in this portfolio under the rolling backtest design. Empirically, the existing accident-period and development components were sufficient to capture the dominant temporal structure in the data. We therefore omit the calendar extension to maintain parsimony and avoid introducing an additional smooth component that does not enhance predictive accuracy.

3.7 Seasonal Effects: Intra-year Cyclicality

Operating at a monthly resolution allows the framework to incorporate seasonal patterns typically obscured in yearly macro-level models. We explored two distinct seasonal components: occurrence seasonality, $\delta_{m(t)}$, and payment seasonality, $\delta_{m(\tau)}$, where $\tau = t + d + s$ denotes calendar month of payment.

The severity predictor was extended to include a payment month effect to capture cyclicality in repair costs or settlement practices:

$$\eta^{\text{sev}}_{t,d,s,j} = \mu^{\text{sev}}_j + \alpha^{\text{sev}}_{j,t} + \beta^{\text{sev}}_{j,d} + \zeta^{\text{sev}}_{j,s} + \delta_{m(\tau)} + \log N^{(j)}_{t,d} \tag{6}$$

where $N^{(j)}_{t,d}$ is the number of claims of type j occurring in accident period t with a reporting delay of d, and $m(\tau)$ maps the calendar period to its corresponding month $m \in \{1, \ldots, 12\}$. To ensure model identifiability, we specify a centered prior $\delta_m \sim \text{Normal}(0, \sigma_m)$.

While occurrence seasonality ($\delta_{m(t)}$) provided limited incremental benefit, the payment month effect ($\delta_{m(t+d+s)}$) improves predictive calibration modestly under the rolling backtest design; quantitative results are reported in Section 5.

As illustrated in Figure 5, the necessity of this parameter is confirmed by the significant relative deviations in weighted average severity across the calendar year. Specifically, the model identifies systemic peaks in Month 10 (exceeding 20% above the global mean) and troughs in Month 6 (10% below the mean). Incorporating $\delta_{m(t+d+s)}$ isolates intra-year cyclicality from the accident-period random walk, preventing seasonal fluctuations from biasing the estimated long-term trends.

3.8 Reporting and Payment Delay Interaction

Standard reserving models typically treat reporting delays and settlement delays as additive and independent components of the claim development process. However, closer inspection of the payment data reveals systematic departures from this assumption. Figure 6 displays the mean log severity across payment delays for different reporting delay groups and reveals pronounced differences in settlement profiles. For claims reported within the first three months, severity is concentrated in early settlement periods and declines rapidly thereafter, reflecting the fact that promptly reported claims tend to involve larger initial settlements that are resolved more quickly. In contrast, claims reported with longer delays exhibit reduced early payments but elevated severities at intermediate

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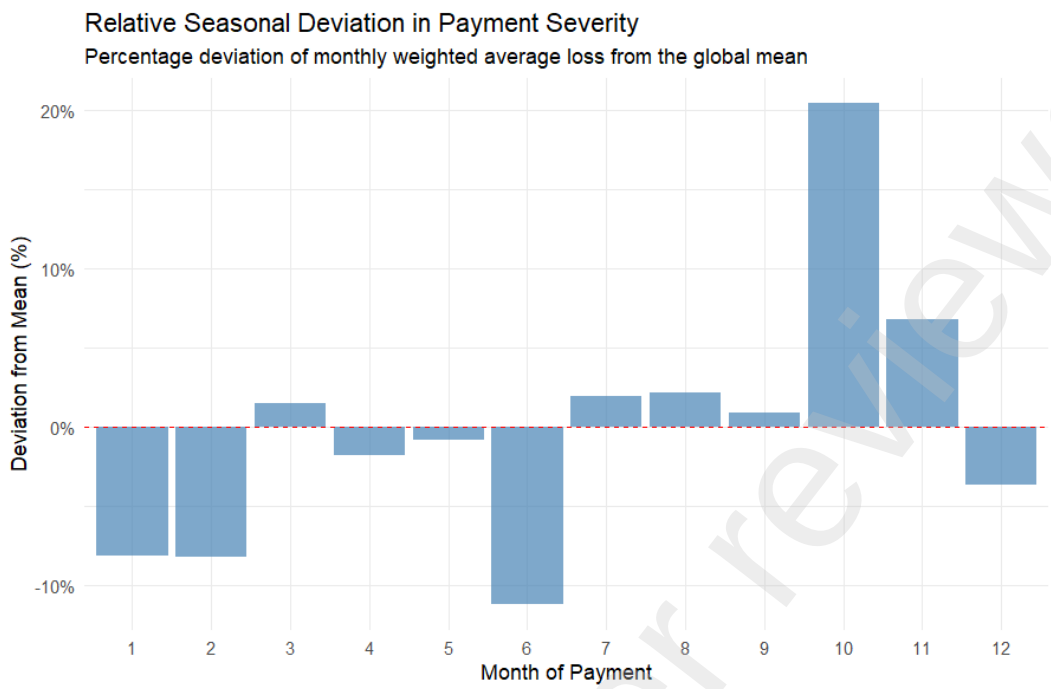

Figure (5) Relative Seasonal Deviation in Payment Severity: Percentage deviation of monthly weighted average loss from the global mean. The substantial deviations in Months 6 and 10 justify the inclusion of the seasonal parameter $\delta_{m(t+d+s)}$.

and late settlement delays, with the decay in severity occurring more gradually. In particular, later-reported claims maintain higher average severity beyond the first year of development relative to promptly reported claims. These systematic shifts in the severity profile across reporting delay groups indicate that reporting delay affects not only the timing of payments but also the evolution of settlement intensity over development. Such patterns cannot be captured by additive reporting and payment delay effects alone, motivating the inclusion of a structured reporting–payment delay interaction in the severity component of the model.

To capture this dependence without introducing the dimensionality and instability associated with a full (d, s) interaction grid, we incorporate a binned interaction structure within the severity component of the model. Reporting delays are grouped into discrete bins $\{0, 1, 2, 3, [4, 6], \geq 7\}$, balancing model flexibility with the pronounced sparsity observed for late-reported claims.

Let $db(d)$ denote the reporting-delay bin to which delay d belongs. The interaction is introduced by allowing each reporting delay bin to induce a distinct modification of the baseline settlement development pattern. Specifically, the severity predictor is extended as

$$\eta^{\text{sev}}_{t,d,s,j} = \cdots + \zeta^{\text{sev}}_{j,s} + \delta_{m(t+d+s)} + \Delta_{db(d),s}, \tag{7}$$

where $\Delta_{db,s}$ represents a bin-specific deviation from the main payment-delay effect.

Each interaction curve is modeled using the same natural spline basis $\mathbf{B}_s$ employed for the settlement effect, such that

$$\Delta_{db,s} = \sigma_\Delta \, \mathbf{B}_s^\top \theta_{db},$$

where θ_{db} is a vector of spline coefficients specific to reporting-delay bin db, and σ_Δ controls the

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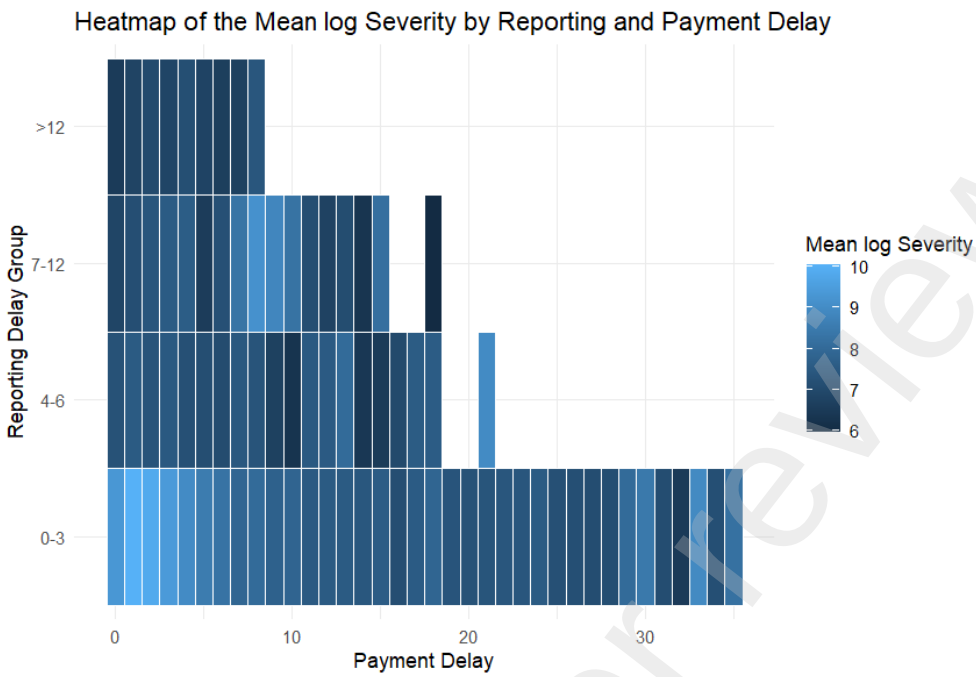

Figure (6) Heatmap of mean log severity by reporting and payment delays. The figure illustrates how the settlement severity profile over payment delay (s) shifts across reporting delay bins (db), suggesting heterogeneous settlement dynamics conditional on reporting speed.

overall magnitude of interaction effects. This representation ensures that the interaction evolves smoothly over settlement delay s, avoiding irregular fluctuations in sparsely observed development periods.

The coefficient vectors satisfy

$$\theta_{db} \sim \text{Normal}(0, 1),$$

and the interaction scale parameter is assigned a regularizing prior

$$\sigma_{\Delta} \sim \text{Half-Normal}(0, 0.5).$$

A slightly tighter scale than that used for the main effects is adopted to reflect the additional flexibility introduced by the interaction term and to mitigate overfitting in sparsely populated reporting-delay bins. A sum-to-zero constraint across settlement delay s is imposed so that $\Delta_{db,s}$ captures deviations in development *shape* only, leaving the overall severity level governed by the main effects.

In practical terms, this specification allows early- and late-reported claims to follow different settlement trajectories over development time without requiring a fully separate severity surface for each reporting delay. High-volume reporting-delay bins are primarily informed by their own experience, while sparse bins are regularized toward the overall development pattern, ensuring stable and actuarially plausible estimation.

Note that in this interaction specification, the main settlement effect ζ_s is modeled entirely through the spline basis. Because the interaction term $\Delta_{db,s}$ already allows reporting-delay groups to modify the early settlement trajectory, retaining separate fixed effects for early settlement periods would

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introduce redundancy between the main and interaction components.

Relative to the additive seasonal specification described in Section 3.7, the reporting–payment interaction improves predictive sharpness by capturing systematic variation in settlement profiles associated with reporting speed. Under the rolling backtest design, this extension yields improved out-of-sample reserve accuracy and tighter predictive intervals without degrading calibration; quantitative comparisons are provided in Section 5.

3.9 Incorporating Portfolio-Level Risk Characteristics

A substantial body of recent literature has explored the use of "micro-level" models for reserving and claims prediction, where individual policyholder and vehicle characteristics are used to explain claim severity and ultimate loss outcomes. These approaches—often based on classical machine learning models, or neural networks—have demonstrated strong predictive performance at the individual claim level (see, for example, Antonio and Plat (2014), Wüthrich (2018a), Wüthrich (2018b), Baudry and Robert (2019), Crevecoeur et al. (2022)).

From a reserving perspective, the primary objective is accurate prediction of aggregate outstanding liabilities at each valuation date. Individual-level models provide valuable evidence that policyholder and vehicle characteristics can be informative for claim costs, and this motivates incorporating risk characteristics into reserving models whenever such information is available. In our setting, we do so in a way that remains consistent with the cell-based structure of the model and with the goal of portfolio-level reserve prediction.

Specifically, rather than modeling individual claims directly, we summarize portfolio composition through cell-level covariates constructed from the cohort of claims contributing to $N_{t,d}^{(j)}$ within each (t, d) cell. For numerical risk characteristics, we compute cell-level averages across this claim cohort, such as the mean driver age, average vehicle age, average engine power, and average vehicle weight. For categorical characteristics, we construct covariates based on empirical proportions within the same cell. In particular, we compute the relative frequencies of vehicle fuel type, vehicle class, geographic region, and renewal status among the claims underlying $N_{t,d}^{(j)}$. This representation allows changes in portfolio mix—rather than individual claim behavior—to influence expected payment patterns, enabling the model to reflect shifts in underlying risk composition within an aggregated reserving framework.

Within the framework, these summary statistics are incorporated as covariates in the severity component of the hurdle model. Let $W_{k,t,d,j}$ denote the standardized value of covariate k computed for claim type j in cell (t, d). We explore two complementary specifications.

The first specification treats the risk characteristics as static drivers of severity. The linear predictor is extended by a linear combination of the standardized covariates:

$$\eta_{t,d,s,j}^{\text{sev}} = \cdots + \sum_{k=1}^{K} \beta_k W_{k,t,d,j}, \tag{8}$$

where β_k represents a global fixed effect for covariate k. We employ a weakly regularizing prior $\beta_k \sim \text{Normal}(0, 0.5)$ to prevent overfitting, consistent with Bayesian practice when handling multiple predictors in sparse environments.

The second specification recognizes that a risk characteristic (e.g., car class) may have a different impact on costs at the beginning of a claim's life than at the end. For instance, high-end vehicle

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classes might influence initial repair costs but have less impact on long-term litigation expenses. To capture this, we allow the covariate coefficients to vary across the payment delay horizon (s):

$$\eta^{\text{sev}}_{t,d,s,j} = \cdots + \sum_{k=1}^{K} \gamma_k(s)\, W_{k,t,d,j}. \tag{9}$$

The function $\gamma_k(s)$ is modeled using a natural spline basis $\mathbf{B}_s$, with each covariate assigned its own spline curve. This allows the model to flexibly capture how the influence of portfolio composition evolves over the settlement horizon. To maintain identifiability, each covariate spline is centered to mean zero, ensuring that the main payment-delay effect remains the primary driver of the overall development shape.

Importantly, both specifications preserve the hierarchical and temporally regularized structure of the baseline model. Covariate effects are therefore estimated jointly with accident-period trends, reporting-delay effects, development patterns, seasonal components, and interaction structures, ensuring that portfolio-level information complements rather than replaces the core reserving dynamics.

For our dataset, incorporating these portfolio-level covariates did not materially improve the prediction of the RBNS reserve. Across the twelve validation dates, both specifications yielded reserve estimates and error metrics that were nearly indistinguishable from those obtained using the temporal model without covariates.

To better understand this behavior, Figure 7 displays the temporal evolution of several representative continuous covariates aggregated at the accident-period level. Because each covariate is computed as an average across many claims within a cell, their variability is substantially attenuated by aggregation. As a result, the series evolve smoothly over time and primarily reflect gradual portfolio drift—for example, increases in average driver or vehicle age due to policy renewals—rather than abrupt structural changes.

Such low-frequency variation is already captured by the accident-period random walk and calendar effects included in the baseline specification. Consequently, the covariates provide limited incremental information beyond what is explained by the structured temporal components. From a reserving perspective, this suggests that the primary drivers of aggregate reserve dynamics in this portfolio are temporal development patterns rather than short-term changes in risk composition. For parsimony, we therefore retain the model without covariates, while noting that this extension may be beneficial in settings with stronger shifts in portfolio mix.

Final Payment specification. Taken together, the empirical evaluation suggests that the most effective and parsimonious specification combines (i) the hurdle Log-normal likelihood, (ii) hierarchical partial pooling across claim types, (iii) seasonal payment effects, and (iv) the reporting–payment delay interaction structure, while omitting calendar period effects and portfolio-level covariates, which provided no material predictive gains for the present dataset. This specification is adopted as the reference payment model for the remainder of the paper. Its out-of-sample performance and uncertainty quantification under the rolling backtest design are reported in Section 5.

3.10 Posterior Computation of the RBNS Reserve

The RBNS reserve at valuation date T is obtained by summing the posterior predictive expectations of future payments for already reported claims. For each posterior draw of model parameters, we

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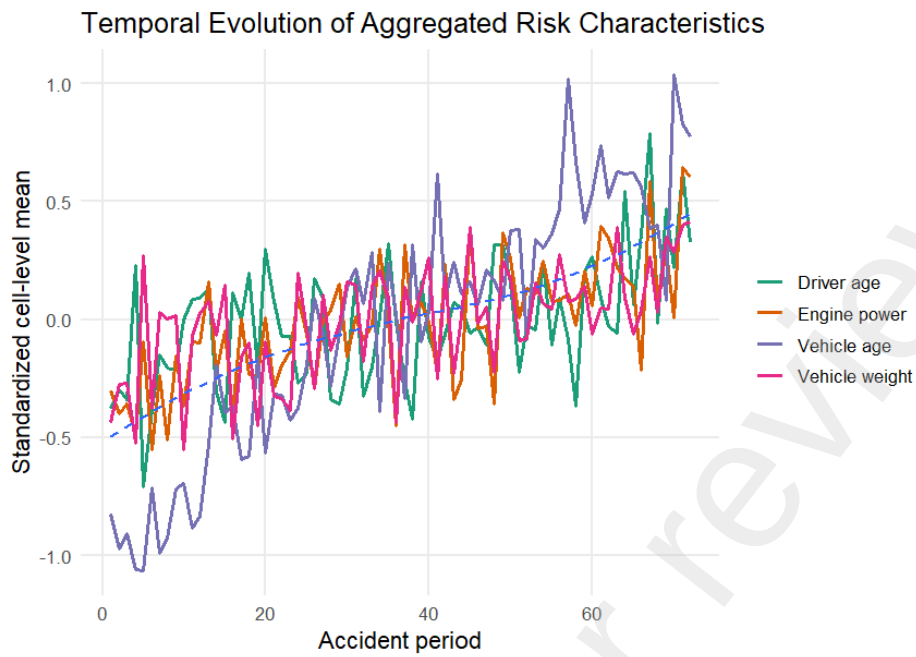

Figure (7) Temporal evolution of selected aggregated continuous portfolio characteristics. Because covariates are computed as cell-level averages, their variation across accident periods is smooth and gradual, providing limited incremental information beyond the temporal effects already captured by the model.

simulate future payment realizations for all cells satisfying $t + d \leq T$ and $t + d + s > T$, and aggregate these simulated payments to obtain a draw from the posterior distribution of the RBNS reserve.

Formally, the reserve is computed as

$$R^{RBNS,(m)} = \sum_{t+d \leq T} \sum_{t+d+s > T} X_{t,d,s}^{(m)},$$

where $X_{t,d,s}^{(m)}$ denotes the simulated future payment under posterior draw m. The posterior distribution of the reserve is then summarized using the median as the point estimate and credible intervals for uncertainty quantification.

4 IBNR Reserving Framework

In this section, we develop a Bayesian framework for estimating IBNR reserves at monthly valuation dates. Unlike the RBNS component described in Section 3, where reported claim counts are fully observed and only the severity model is required, the IBNR problem involves claims that have occurred but have not yet been reported. Reserve estimation therefore requires an additional modeling layer to predict unreported claim counts, which are subsequently propagated through the severity model to generate future payments.

Consistent with the empirical characteristics documented in Section 2, the proposed framework is designed to capture the highly front-loaded reporting delay distribution and substantial over-

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dispersion in claim frequency relative to exposure. Methodologically, the count component builds on the framework developed in Abdelrahman et al. (2025), with extensions that stabilize inference in sparsely populated long-delay regions and enable coherent allocation of predicted counts across claim types. Together, these components provide stable and interpretable inputs to the downstream reserve calculation.

4.1 IBNR Definition

Let $N_{t,d}$ denote the number of claims occurring in accident month t and first reported with delay d. At valuation date T, observations are available only for cells satisfying $t + d \leq T$, while cells with $t + d > T$ correspond to claims that have occurred but have not yet been reported. These unobserved cells constitute the IBNR portion of the development triangle.

The total IBNR claim count is defined as

$$\hat{C}^{IBNR} = \sum_{t+d>T} \mathbb{E}[N_{t,d} \mid \mathcal{D}_T]\,, \tag{10}$$

where $\mathcal{D}_T$ denotes all information available at time T. Predicted counts serve as exposures in the severity model described in Section 3, enabling the estimation of the associated outstanding payments.

4.2 Claim Count Model

We model the observed cell-level claim counts $N_{t,d}$ for $t+d \leq T$ using a Negative Binomial regression to accommodate over-dispersion:

$$N_{t,d} \,\big|\, \lambda_{t,d}, \phi \;\sim\; \text{Negative Binomial}(e_t \lambda_{t,d}, \phi), \qquad \mathbb{E}[N_{t,d} \mid \lambda_{t,d}] = \lambda_{t,d}, \tag{11}$$

where e_t represents an exposure offset, ensuring that the model accounts for variations in the number of insureds in accident period t, and ϕ is the overdispersion parameter. The conditional mean is specified on the log scale as

$$\log \lambda_{t,d} = \mu + \alpha_t + \beta_d + \mathbf{x}_t^\top \boldsymbol{\delta}_d + \mathbf{1}_{\{d>k_0\}} \tau_t + \log(\mathrm{e}_t), \tag{12}$$

where μ is an intercept, α_t is an accident-period effect, β_d is a reporting-delay effect, and e_t is an offset. The indicator term introduces an additional *tail-only* accident-period component τ_t that acts only for delays exceeding a threshold k_0, defined such that approximately 99.9% of claims are reported by $d \leq k_0$.

Accident-period effects. To capture gradual evolution in claim frequency while stabilizing estimates across months, we model α_t using a first-order random walk,

$$\alpha_t = \alpha_{t-1} + \sigma_\alpha \, \varepsilon_t, \qquad \varepsilon_t \sim \text{Normal}(0, 1),$$

with a zero-mean constraint for identifiability.

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Reporting delay effects and tail regularization. The reporting delay distribution is strongly front-loaded and sparse in the tail. We therefore adopt a hybrid representation of β_d:

$$\beta_d = \begin{cases} \beta_d^{(\text{fixed})}, & d \le k_0, \\ \beta_{k_0}^{(\text{fixed})} + \mathbf{B}(d)^\top \mathbf{b}, & d > k_0, \end{cases}$$

where $\mathbf{B}(d)$ is a low-dimensional spline basis and $\mathbf{b}$ are spline coefficients. This structure preserves flexibility where data are abundant ($d \le k_0$) while enforcing smoothness in sparsely observed long delays.

Tail-only accident-period interaction. During backtesting, we observed that the implied contribution of long reporting delays to IBNR increased in the months preceding certain valuation dates. To accommodate this behavior without distorting the well-identified early delays, we include a tail-only term τ_t in (12) for $d > k_0$. We model τ_t as a first-order random walk:

$$\tau_t = \tau_{t-1} + \sigma_\tau \, \xi_t, \qquad \xi_t \sim \text{Normal}(0, 1),$$

again with a centering constraint.

Calendar covariates. The vector $\mathbf{x}_t$ contains calendar-related covariates describing the month of occurrence and the final day of the month. These variables influence the reporting delay structure through seasonal reporting dynamics and operational end-of-month effects. Allowing the coefficient vector $\boldsymbol{\delta}_d$ to vary by delay d enables these covariates to act directly on the delay profile. Further discussion and empirical motivation are provided in Abdelrahman et al. (2025)."

4.3 Claim-Type Decomposition with a Nowcast Anchor

Let $N_{t,d}^{(j)}$ denote the number of claims of type $j \in \{1, \ldots, J\}$ in cell (t, d), with

$$\sum_{j=1}^{J} N_{t,d}^{(j)} = N_{t,d}.$$

Because the severity model in Section 3 is specified hierarchically by claim type, we require coherent type-level IBNR count predictions. Conditional on the aggregate count, we model the type composition as

$$\left(N_{t,d}^{(1)}, \ldots, N_{t,d}^{(J)}\right) \mid N_{t,d}, \boldsymbol{\pi}_{t,d} \;\sim\; \text{Multinomial}(N_{t,d}, \boldsymbol{\pi}_{t,d}) \,. \tag{13}$$

Baseline composition anchored at $d = 0$. To ensure stable identification of the composition dynamics, we anchor the baseline composition at reporting delay $d = 0$, which is observed at the time of reporting for each accident period. This establishes a reference distribution at the earliest reporting stage, allowing subsequent reporting-delay effects to be interpreted as deviations from this baseline.

Let $\boldsymbol{\pi}_{t,0}$ denote the baseline type mix in accident month t. Using type 1 as a reference, define baseline log-odds $\eta_{t,j}$ for $j = 2, \ldots, J$ and set

$$\boldsymbol{\pi}_{t,0} = \text{softmax}(0, \eta_{t,2}, \ldots, \eta_{t,J}) \,.$$

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We model each $\eta_{t,j}$ as a first-order random walk,

$$\eta_{t,j} = \eta_{t-1,j} + \sigma_{\eta,j} u_{t,j}, \qquad u_{t,j} \sim \text{Normal}(0,1),$$

with centering constraints for identifiability. The random walk prior stabilizes estimation in low-volume months while allowing rapid adaptation when the observed $d = 0$ composition exhibits a genuine structural shift.

Delay-driven selection as a relative tilt. While the dominant composition signal is captured by $\boldsymbol{\pi}_{t,0}$, reporting delay may induce mild selection effects (e.g., later-reported claims being enriched for particular types). We represent this through an additive *delay tilt* in logit space:

$$\boldsymbol{\pi}_{t,d} = \text{softmax}(0, \eta_{t,2} + g_{d,2}, \ldots, \eta_{t,J} + g_{d,J}), \qquad g_{0,j} = 0.$$

This formulation separates two interpretable mechanisms: (i) $\eta_{t,j}$ captures *portfolio mix changes over accident time* (nowcasted through $d = 0$), and (ii) $g_{d,j}$ captures a *stable reporting-delay selection profile* shared across accident months.

Piecewise structure in the delay tilt. Because delays beyond k_0 are extremely sparse, estimating a fully free $g_{d,j}$ for all d is unstable. We therefore adopt a piecewise specification:

$$g_{d,j} = \begin{cases} 0, & d = 0, \\ g_{d,j}^{(\text{body})}, & 1 \le d \le k_0, \\ g_j^{(\text{tail})}, & d > k_0, \end{cases}$$

so that the model can learn distinct tilts across the well-populated early delays while pooling all long delays into a single tail tilt for stability.

4.4 Prior Structure

To ensure stable inference across the sparse (t, d) triangle, we employ weakly informative priors with non-centered parameterizations.

Total counts. Random walk innovations for α_t and the tail trend τ_t follow standard Normal distributions, with Half-Normal priors on the corresponding smoothness scales σ_α and σ_τ. Early delay effects $\beta_d^{(\text{fixed})}$ are assigned zero-centered Normal priors, while spline coefficients $\mathbf{b}$ in (4.2) follow Normal priors with an additional shrinkage scale to regularize the tail. The over-dispersion parameter ϕ is assigned a weakly informative prior to permit substantial extra-Poisson variability.

Composition. Baseline log-odds innovations $u_{t,j}$ in (4.3) are standard Normal, with type-specific Half-Normal scales $\sigma_{\eta,j}$ controlling smoothness over accident time. Body delay tilts $g_{d,j}^{(\text{body})}$ are assigned centered Normal priors with strong shrinkage toward zero, reflecting the assumption that delay selection is present but modest. Tail tilts $g_j^{(\text{tail})}$ receive tighter zero-centered Normal priors, enforcing pooling and stability in the sparse long-delay region.

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4.5 Reserve Generation

Given posterior samples of the count and composition models, IBNR reserves are generated through a posterior predictive simulation procedure consistent with the RBNS severity framework. For each posterior draw, predicted counts $N_{t,d}^{(j)}$ are generated for all cells with $t + d > T$. These predicted counts enter the severity model as exposures through the offset $\log N_{t,d}^{(j)}$, and future payments $X_{t,d,s}^{(j)}$ are simulated for all cells satisfying $t + d + s > T$ using the hurdle Log-normal specification.

The aggregate IBNR reserve is computed as

$$\hat{R}^{IBNR} = \sum_{t+d>T} \sum_{t+d+s>T} \sum_{j=1}^{J} X_{t,d,s}^{(j)},$$

and repeating this procedure across posterior samples yields the predictive distribution of the reserve.

To reduce excess Monte Carlo variability induced by heavy-tailed severity, we use posterior median predicted counts within each (t, d, j) cell as fixed exposures when simulating payments. Consistent with the RBNS framework, the posterior median of the simulated reserve distribution is adopted as the point estimate.

4.6 Model Validation

The IBNR framework is evaluated using the same rolling backtest procedure adopted for the RBNS component. For each of twelve monthly valuation dates, the model is fitted using only information available at time T, and reserves are predicted for the unobserved portion of the development triangle. Predictive performance is assessed by comparing posterior predictive summaries with subsequently observed claim reports and payments.

5 Results and Predictive Performance

In this section, we evaluate the predictive performance of the proposed Bayesian framework for estimating RBNS and IBNR reserves. All models are assessed using the rolling-origin backtesting design described in Section 3. In brief, for each of twelve monthly valuation dates in 2014, models are fitted using only information available at the valuation date. Posterior samples obtained via `Stan` are propagated through the predictive mechanisms of the framework to generate posterior predictive distributions of RBNS reserves, IBNR claim counts, and IBNR reserves. These components are then aggregated to obtain the predictive distribution of the total reserve. Predictive accuracy is evaluated using the Mean Absolute Percentage Error (MAPE) together with posterior predictive intervals. Given the heavy-tailed nature of insurance payments, posterior medians are used as point estimates to ensure robust reserve predictions.

5.1 RBNS Reserve Performance

We first evaluate the predictive performance of the RBNS severity framework using the sequential specifications introduced in Section 3. Starting from a parsimonious baseline, additional structural components are incorporated progressively to assess their incremental impact on predictive accuracy and stability.

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Table 4 summarizes the out-of-sample performance of four specifications:

- Model A: baseline hurdle Log-normal model,
- Model B: Model A with hierarchical partial pooling across claim types,
- Model C: Model B with seasonal payment effects,
- Model D: Model C with reporting–payment delay interaction.

Table (4) Out-of-sample RBNS predictive performance across twelve valuation dates. Performance is measured using the Mean Absolute Percentage Error (MAPE) based on posterior median predictions.

Model	Description	MAPE (%)
A	Baseline hurdle Log-normal	14.7
B	+ Hierarchical pooling across claim types	7.5
C	+ Seasonal payment effects	7.1
D	+ Reporting–payment interaction	6.4

Introducing hierarchical structure across claim types (Model B) produces the largest improvement, reducing the MAPE by nearly half relative to the baseline. This indicates that heterogeneity in severity profiles across claim causes is a primary driver of prediction error and that partial pooling effectively stabilizes inference in low-volume segments. Adding seasonal effects (Model C) yields a modest additional gain by capturing intra-year cyclicality in settlement behavior. Finally, incorporating the reporting–payment delay interaction (Model D) further improves accuracy, reflecting the presence of systematic differences in settlement trajectories conditional on reporting speed. Calendar-period effects and portfolio covariates are omitted, as they did not provide measurable predictive gains in the backtest.

Beyond improvements in point accuracy, the interaction structure also enhances predictive sharpness. As illustrated in Figure 8, the final specification produces tighter predictive intervals while maintaining full empirical coverage of observed outcomes. This indicates that the additional structure captures systematic variation that would otherwise appear as residual noise. From a reserving perspective, this increased precision supports more reliable reserve estimation and improved assessment of capital adequacy.

5.2 IBNR Count and Reserve Performance

We next evaluate the predictive performance of the IBNR component, considering both the aggregate IBNR claim count and the resulting IBNR reserve.

IBNR Claim Count. The count model exhibits strong and stable predictive accuracy across all twelve valuation dates. Posterior median forecasts yield a mean MAPE of 7.3%, and the associated 90% posterior predictive intervals cover the realized claim counts at every valuation date. Figure 9 displays the predictive intervals together with the observed outcomes, illustrating both low bias and well-calibrated uncertainty.

These findings are consistent with the results reported in Abdelrahman et al. (2025), where the same Bayesian count framework was shown to outperform a wide range of stochastic micro-level frequency models as well as the traditional chain ladder method when applied to this dataset.

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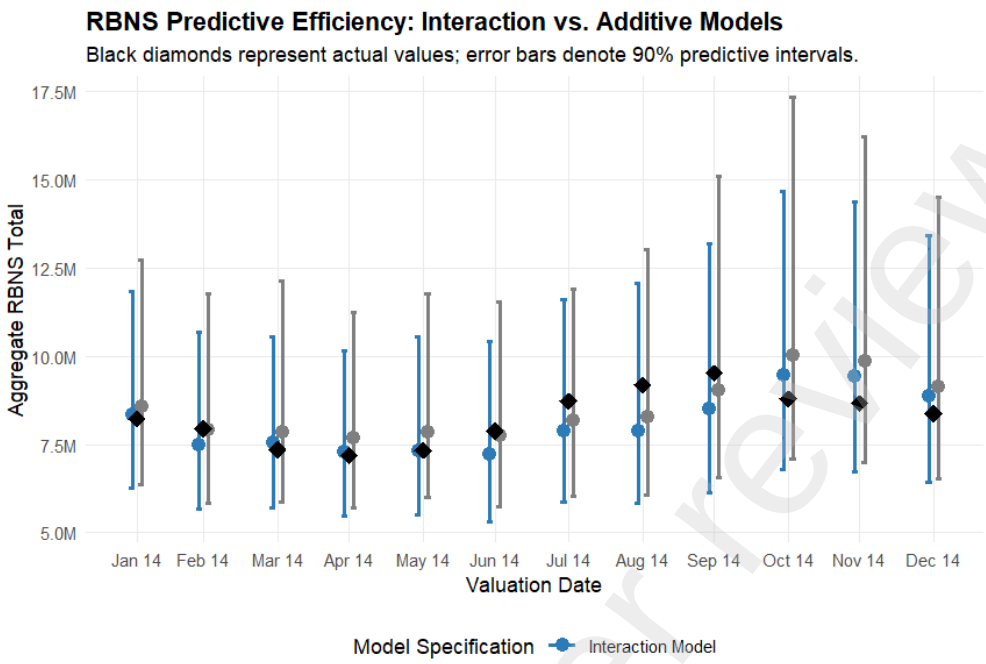

Figure (8) Out-of-sample RBNS predictive performance: comparison of the interaction model (blue) against the additive seasonal model (gray). The interaction framework yields tighter 90% predictive intervals and improved accuracy while maintaining full empirical coverage of actual outcomes (black diamonds).

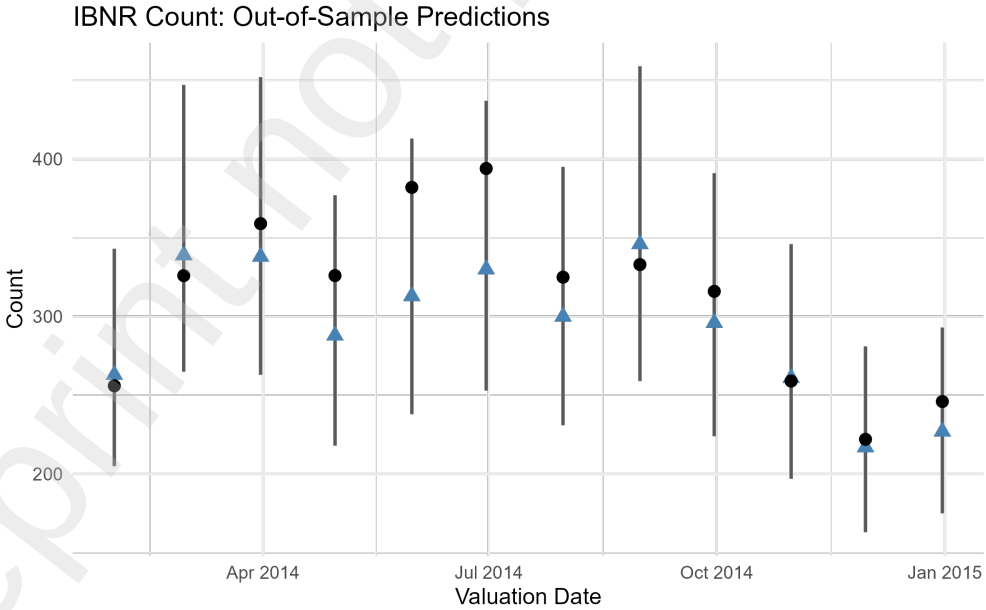

Figure (9) Out-of-sample predictive performance for the aggregate IBNR claim count across the twelve valuation dates. Black circles denote realized counts, blue triangles represent posterior median predictions, and vertical bars indicate 90% posterior predictive intervals.

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IBNR Reserve. Predictive accuracy for the IBNR reserve is naturally lower, with a MAPE of 18.3%. This increase in error is expected and reflects the multiplicative structure of the joint framework. Reserve projections compound uncertainty from two sources: (i) the prediction of future claim counts and their allocation across claim types, and (ii) the simulation of heavy-tailed payment severities conditional on those exposures. Small deviations in reporting delay distributions or claim-type composition are naturally amplified when propagated through the log-linear severity layer.

Despite the higher relative error, calibration remains strong. For all valuation dates, realized IBNR reserves lie within the 5th and 95th percentiles of the posterior predictive distribution, indicating that the framework appropriately captures the underlying process uncertainty. Figure 10 illustrates the predictive intervals and observed reserves over time.

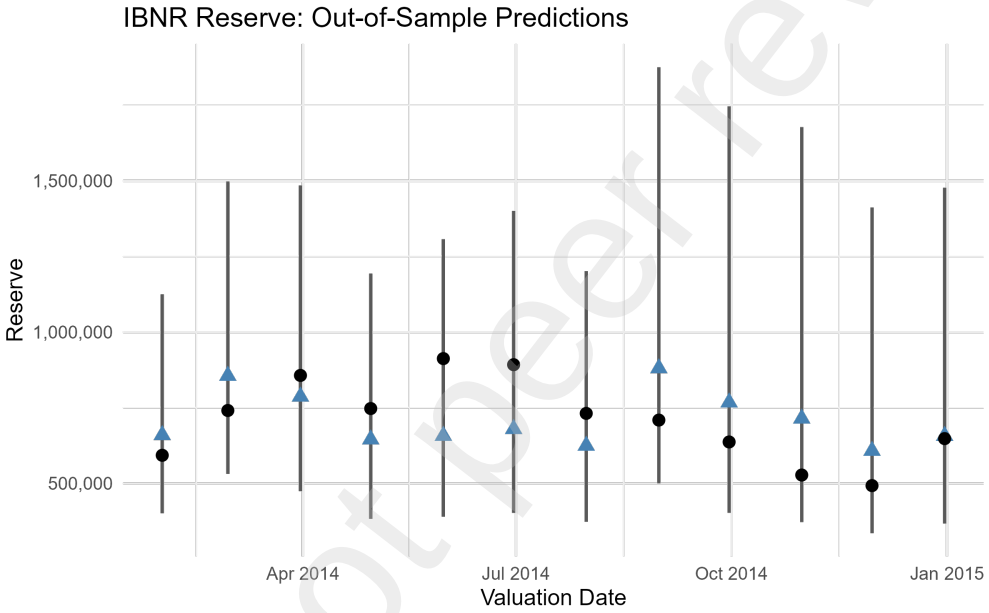

Figure (10) Out-of-sample predictive performance for the aggregate IBNR reserve across the twelve valuation dates. Black circles denote realized reserves, blue triangles represent posterior median predictions, and vertical bars indicate 90% posterior predictive intervals.

Importantly, the higher variability in the IBNR reserve has limited impact on the aggregate portfolio reserve. Within this portfolio, the RBNS component constitutes the majority of outstanding liabilities, while the IBNR portion represents a smaller share of the total exposure. Consequently, even moderate relative deviations in the IBNR reserve translate into comparatively small absolute differences in the overall reserve estimate. This explains why the combined total reserve maintains substantially lower aggregate prediction error, as discussed in the following subsection.

5.3 Total Reserve Performance

The final evaluation concerns the predictive performance of the aggregate total reserve, defined as the sum of RBNS and IBNR liabilities. This portfolio-level measure is most relevant for assessing overall reserve adequacy and capital planning.

As illustrated in Figure 11, the joint Bayesian framework demonstrates high predictive stability across the twelve valuation dates. The total reserve estimate achieves a MAPE of 6.3%. Notably,

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this aggregate error is lower than the standalone RBNS error (6.4%), which can be attributed to a diversification effect where slight overestimations in one component are naturally offset by underestimations in the other.

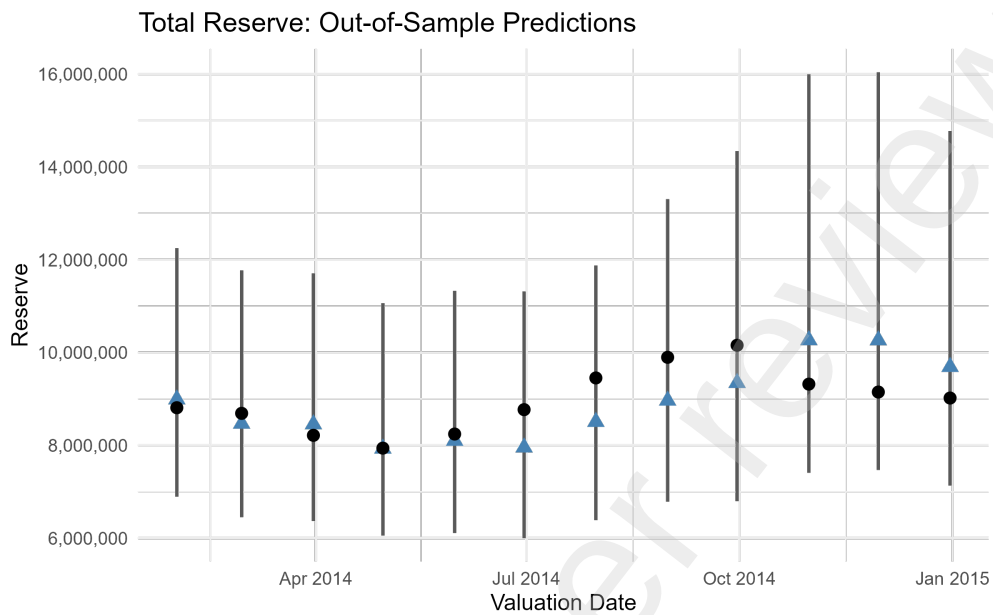

Figure (11) Out-of-sample predictive performance for the aggregate total reserve across the twelve valuation dates. Black circles denote realized reserves, blue triangles represent posterior median predictions, and vertical bars indicate 90% posterior predictive intervals.

To contextualize the performance of the proposed model, we compare the results against several specifications of the Mack Chain Ladder (CL), the prevailing industry standard for portfolio-level reserving. We evaluate the CL across two dimensions: temporal granularity (Yearly vs. Monthly) and structural decomposition (Aggregate vs. Decomposed by claim type).

The comparative results are summarized in Table 5.

Table (5) Comparison of Total Reserve predictive performance (MAPE %) across twelve valuation dates.

Model Specification	Granularity	Structure	MAPE (%)
Joint Bayesian Framework	**Monthly**	**Decomposed**	**6.3**
Chain Ladder (Benchmark A)	Yearly	Aggregate	16.2
Chain Ladder (Benchmark B)	Yearly	Decomposed	16.1
Chain Ladder (Benchmark C)	Monthly	Aggregate	47.2
Chain Ladder (Benchmark D)	Monthly	Decomposed	48.7

The benchmarking results reveal several critical insights regarding the limitations of deterministic methods:

- **Granularity and Stability:** While the yearly Chain Ladder specifications (Benchmarks A and B) provide relatively stable predictions with MAPEs around 16%, they are inherently limited by their inability to capture intra-year signals. We also implemented the Chain Ladder using monthly aggregation to match the granularity of our proposed model; however, these specifications (Benchmarks C and D) proved highly unstable, with errors exceeding 47%. This "granu-

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larity penalty" arises because deterministic factors over-react to sparse data cells and zero-heavy reporting tails in monthly triangles, leading to a "multiplier explosion" in the projected reserve.

- **The Benefit of Decomposition:** In the yearly Chain Ladder, decomposing the data by claim type (Benchmark B) provided only a marginal improvement over the aggregate approach (Benchmark A). This suggests that without a formal mechanism to handle heterogeneity, simply splitting the data provides limited value for our application.
- **The Bayesian Advantage:** Our joint Bayesian framework effectively bridges these gaps. By utilizing hierarchical partial pooling, the model "borrows strength" across claim types to stabilize sparse segments while simultaneously capturing high-resolution monthly signals (such as the identified frequency spikes in May and June) through Random Walk priors.

The fact that the proposed model maintains a 6.3% error at a monthly scale—significantly outperforming even the more stable yearly Chain Ladder—demonstrates the superior ability of the Bayesian framework to extract meaningful signals from high-frequency insurance data without succumbing to the volatility that plagues traditional methods.

Implementation details and reproducibility guidelines are provided in Appendix A, which documents the full `Stan` workflow, including model compilation using `cmdstanr` and posterior reserve simulation. The appendix also references the public repository containing the `Stan` implementation used in this study.

6 Conclusion

This paper develops a unified Bayesian framework for reserving that builds on the CRM formulation of Wahl et al. (2019) and jointly models RBNS and IBNR liabilities within a single coherent probabilistic structure. The proposed approach combines a severity component for reported claims with a frequency model for unreported claims, linking the two layers through a common exposure formulation. This design enables both parts of the reserve to be estimated consistently while fully propagating parameter and process uncertainty through posterior predictive simulation.

Rather than proposing a single fixed specification, the framework demonstrates how a range of structural components commonly encountered in reserving practice can be incorporated within a unified Bayesian model. These include hierarchical pooling across claim types, smooth temporal dynamics, development effects, seasonal patterns, calendar-year effect, and interactions across reporting and payment delays. This modular structure allows such features to be introduced or omitted in a principled manner depending on data characteristics, providing flexibility while maintaining stability in sparse triangles. All components are estimated jointly using Hamiltonian Monte Carlo in `Stan`, illustrating how modern probabilistic programming tools enable complex multi-layer reserving models to be implemented transparently within a single likelihood.

Empirical evaluation using a rolling backtest across twelve monthly valuation dates shows that the proposed framework delivers substantial improvements in predictive accuracy relative to traditional deterministic methods. At the aggregate level, the total reserve achieves a mean absolute percentage error of approximately 6.3%, materially outperforming standard annual Chain Ladder benchmarks. These results demonstrate that our framework can effectively extract high-frequency information while avoiding the instability that often arises when deterministic techniques are applied to aggregate run-off triangles.

From a practical perspective, the framework provides several advantages. It produces calibrated

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predictive intervals that directly quantify reserve uncertainty, supporting risk assessment and capital planning. Hierarchical pooling stabilizes sparse segments without sacrificing interpretability, and the modular design allows practitioners to tailor model complexity to their portfolio. To facilitate adoption, full implementation details and `Stan` code for the final specifications are provided in the Appendix, enabling both practitioners and researchers to readily apply and extend the methodology to their own datasets.

Overall, the results indicate that structured Bayesian collective reserving offers a principled, flexible, and practically effective alternative to traditional approaches, delivering improved predictive performance together with coherent uncertainty quantification.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process During the preparation of this work, the authors used ChatGPT (OpenAI) to assist in improving the clarity and language of the manuscript, as well as for general support with data engineering tasks and code writing. After using this tool, the authors carefully reviewed and edited the generated content as needed and take full responsibility for the content of the article.

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A Implementation Guide

This appendix documents (i) how to compile and fit the joint count–composition–severity model in `Stan` using `cmdstanr`, and (ii) how to generate posterior predictive distributions of RBNS, IBNR, and total reserves from saved posterior draws. The objective is to document the practical implementation of the proposed framework by separating the workflow into modular components: model fitting in `Stan`, followed by reserve simulation as an external post-processing step. The `Stan` code used in this paper is publicly available at `https://github.com/HassanAbdelrahman/Bayesian-Reserving-Model`.

Data engineering is required to transform raw claim, payment, and exposure records into the structured inputs described below. These transformations are portfolio-specific (e.g., claim-type defini-

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tions, delay caps, and binning rules) and depend on the structure of the underlying administrative data. Because the dataset used in this study is proprietary, we do not provide a dataset-specific data engineering pipeline. Instead, the repository provides the `Stan` implementation of the model, illustrating how the proposed probabilistic framework can be applied once the required aggregated inputs have been constructed.

A.1 Fitting the Joint Model in `cmdstanr`

The joint model is implemented in the file `joint_count_severity.stan`. It combines: (i) a negative binomial model for total reported counts by accident period and reporting delay, (ii) a multinomial model for claim-type composition conditional on totals, and (iii) a hurdle–lognormal model for payment occurrence and payment size.

The model can be fit as follows:

```
library(cmdstanr)

stan_file <- "joint_count_severity.stan"
mod <- cmdstan_model(stan_file)

fit <- mod$sample(
  data = stan_data_joint,
  chains = 4,
  parallel_chains = 4,
  seed = 123,
  refresh = 0
)
```

A.2 Stan Data Inputs: Structure and Interpretation

The `Stan` program expects the data in *long format*: each observation is represented by a row index, while the corresponding accident period, delay, settlement index, and claim type are provided via parallel index arrays. Accident periods are indexed by $t = 1, \ldots, L$. Reporting delays are indexed in `Stan` as $d = 1, \ldots, D$, where $d = 1$ corresponds to calendar delay $d_0 = 0$.

A.2.1 Count component (totals and composition)

The count likelihood is evaluated on $N_{\text{obs,count}}$ observed (t, d) cells. Each row index $n \in \{1, \ldots, N_{\text{obs,count}}\}$ corresponds to one observed cell

$$(t_n, d_n) = \big(\texttt{time_count[n]},\ \texttt{delay_count[n]}\big),$$

and the data contain the total count `y_tot[n]` along with the type-specific counts `y_type[n,1:J]` for that same cell.

Dimensions.

- `L`: number of accident periods.
- `D`: number of delay indices in `Stan` (so $D = D_0 + 1$ if calendar delays are $d_0 = 0, \ldots, D_0$).

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- `p`: number of period-level covariates.
- `J`: number of claim types.

Observed cell rows (likelihood).

- `N_obs_count`: number of observed count cells (rows).
- `time_count[n]`: accident period index $t_n \in \{1, \ldots, L\}$ for row n.
- `delay_count[n]`: delay index $d_n \in \{1, \ldots, D\}$ for row n (with $d_n = 1$ meaning calendar delay 0).
- `y_tot[n]`: observed total count in cell (t_n, d_n).
- `y_type[n,j]`: observed type-specific counts for $j = 1, \ldots, J$ in cell (t_n, d_n).
- `exposure_count[n]`: exposure associated with cell (t_n, d_n) (used as an offset).
- `covariates[t,k]`: period-level covariates for $t = 1, \ldots, L$ and $k = 1, \ldots, p$.

Piecewise delay structure.

- `k0`: body/tail cutoff on the `Stan` delay index; delays $d \leq$ `k0` are treated as the body.
- `K_spline`, `B_spline`: spline basis used to parameterize the tail part of the delay effect.

A.2.2 Severity component (payment occurrence and amount)

The severity likelihood is evaluated on $N_{\text{obs,sev}}$ observed payment rows. Each row index $n \in \{1, \ldots, N_{\text{obs,sev}}\}$ corresponds to one aggregated payment cell indexed by (t, d, s, j):

$$(t_n, d_n, s_n, j_n) = \big(\texttt{t_obs[n]}, \texttt{d_obs[n]}, \texttt{s_obs[n]}, \texttt{j_obs[n]}\big),$$

where s denotes the settlement index (e.g., payment delay in months), and j denotes claim type.

Dimensions and spline basis.

- `T_max`, `D_max`, `S_max`, `J_max`: maximum indices used by the severity model for accident period, reporting-delay index, settlement-delay index, and claim type.
- `K_zeta`, `B_zeta_full`: spline basis over the settlement index used in the severity predictors.

Observed payment rows (likelihood).

- `N_obs_sev`: number of observed payment rows (cells).
- `t_obs[n]`, `d_obs[n]`, `s_obs[n]`, `j_obs[n]`: indices (t_n, d_n, s_n, j_n) for row n.
- `P_obs[n]`: aggregated payment amount.
- `Z_obs[n]`: indicator that the payment amount is positive.
- `log_N_obs[n]`: log-count offset included in severity predictors.

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- `d_bin_obs[n]`: delay-bin category (values $1, \ldots, 6$).
- `month_sev[n]`: calendar month indicator (values $1, \ldots, 12$).

A.3 Stan Program

The `Stan` implementation used in our empirical study is available in the public repository referenced above as `joint_count_severity.stan`. The program corresponds to the final joint specification described in Sections 3 and 4, including the priors and identifiability constraints used in the reported results.

A.4 Saving a Compact Posterior Draw Object

To decouple reserve simulation from model fitting, we save a compact draw object containing only the parameters needed for posterior predictive reserving, which reduces storage and improves post-processing speed:

```
library(posterior)

vars_needed <- c(
  # count component
  "mu","alpha","beta","delta","sigma_nb","tail_eff","eta","g",
  # severity component
  "mu_p_prob","mu_p_sev",
  "beta_p_prob","beta_p_sev",
  "alpha_p_prob","alpha_p_sev",
  "zeta_p_prob","zeta_p_sev",
  "delta_month_sev","sigma_sev","theta_log_n_prob",
  # ingredients for reconstructing delay-bin x settlement interaction
  "c_zeta_sev_d_0","sigma_c_zeta_sev_d","c_zeta_sev_d_raw"
)

draws_core <- fit$draws(variables = vars_needed, format = "draws_df")
saveRDS(draws_core, "draws_core.rds")
```

A.5 Posterior Predictive Reserve Simulation

Reserve distributions are obtained by simulating from the posterior predictive distribution conditional on posterior draws of the joint model parameters.

Because the dataset used in this study is proprietary, and several grid-construction and modeling choices are portfolio-specific, we present here a general implementation algorithm rather than portfolio-dependent code.

Inputs. Reserve simulation requires:

- posterior draws of the transformed parameters from the fitted `Stan` model;
- a prediction set of unobserved (t, d) count cells (IBNR region);
- a payment prediction grid for post-valuation payments, containing both RBNS and IBNR rows;

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- exposure and covariate matrices aligned with accident-period index.

Notation. Let T denote the final observed accident-period index at the valuation date. Unobserved count cells satisfy $t + d > T$.

Algorithm 1 Stabilized Posterior Predictive Reserve Simulation

Require: All posterior draws S, count prediction cells, payment prediction grid, covariates, and exposures.

Ensure: Posterior predictive distributions of RBNS, IBNR, and total reserves.

Step 1: Frequency Simulation (All Draws)
for each draw $s = 1, \ldots, S$ **do**
  Simulate total IBNR counts $N_{t,d}^{(s)}$ and type-specific counts $N_{t,d,j}^{(s)}$ using the count and composition modules.
end for
Step 2: Construct Stabilized Frequency Inputs
Compute the cell-wise posterior median $\hat{N}_{t,d,j} = \text{median}(\{N_{t,d,j}^{(s)}\}_{s=1}^{S})$ for all IBNR cells.
Step 3: Severity Simulation (Per Draw)
for each draw $s = 1, \ldots, S$ **do**
  Assign log-count offsets:
  - For RBNS rows, use observed reported counts from the valuation data.
  - For IBNR rows, use the stabilized median $\log(\hat{N}_{t,d,j})$.

  Simulate payment occurrence and size using severity parameters from draw s.
  Aggregate to obtain $\text{RBNS}^{(s)}$, $\text{IBNR}^{(s)}$, and $\text{Total}^{(s)}$.
end for

Repeating Algorithm 1 for $s = 1, \ldots, S$ yields the posterior predictive distributions of RBNS, IBNR, and total reserves.

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